

Report 1: Research Motivation Categories

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Part 1: Category Summary

Category	Count	Percentage
Disease-Motivated Research	12	44.4%
Basic Metabolic Biology	5	18.5%
Basic Cellular Biology	9	33.3%
Other	1	3.7%

Disease-Motivated Research (12 posters)

Definition: Research explicitly motivated by understanding, diagnosing, or treating specific human diseases.

Examples: Studies on autism, Rett syndrome, Parkinson's, cancer, cardiovascular disorders.

Posters:

- Avetisyan et al. (2023) - CCC SP23
- Brown et al. (2025) - CCC SP25
- Cabral et al. (2025) - CCC FA25
- Gill & Alcazar (2025) - CCC FA25
- Haubelt & Alcazar et al. (2025) - CCC FA25
- Logan et al. (2025) - CCC FA25
- Meraz et al. (2023) - CCC SP23
- Paderna et al. (2025) - CCC SP25
- Paramo-Ojeda et al. (2025) - CCC SP25
- Rodriguez (2025) - CCC SP25
- Trevino et al. (2022) - CCC SP22
- Zlaket et al. (2023) - CCC SP23

Basic Metabolic Biology (5 posters)

Definition: Research focused on understanding metabolic processes, nutrient metabolism, and energy homeostasis.

Examples: Studies of carbohydrate metabolism, fatty acid synthesis, energy homeostasis genes.

Posters:

- Berbouti et al. (2025) - CCC FA25
- Camara et al. (2025) - LU SP25
- Dehuelbes et al. (2025) - LU SP5
- Nii et al. (2025) - CCC FA25
- Ojala et al. (2025) - LU SP25

Basic Cellular Biology (9 posters)

Definition: Research focused on fundamental cellular processes and physiological functions.

Examples: Studies of immune function, digestion, aging, neuroendocrine signaling.

Posters:

- Diaz et al. (2025) - CCC SP25
- Ford et al. (2023) - CCC SP23
- Godoy-Pena et al. (2025) - CCC FA25
- Henriquez et al. (2025) - COD WI24
- Holmes (2025) - CCC SP25
- Lemus et al. (2025) - FA25 CCC
- Luera et al. (2025) - CCC FA25
- Pedireddi et al. (2025) - CCC SP25
- Tuttle et al. (2025) - CCC SP25

Other (1 posters)

Definition: Research that does not clearly fit into the above categories.

Examples: Studies of developmental patterning, tissue organization.

Posters:

- Sakana et al. (2025) - CCC SP25

Part 2: Individual Poster Evidence

Avetisyan et al. (2023) - CCC SP23

Category: Disease-Motivated Research

- Research focused on autism-related genes
- Introduction discusses gut-microbiome-brain axis and Autism
- Goal: research autism genes in *D. Melanogaster* midgut

Berbouti et al. (2025) - CCC FA25

Category: Basic Metabolic Biology

- Primary focus on trehalose catabolism
- Investigates differential gene expression
- No disease focus

Brown et al. (2025) - CCC SP25

Category: Disease-Motivated Research

- Title mentions Acid Reflux Related Genes
- Introduction: acid reflux affects millions
- Goal: understanding and treatment of acid reflux

Cabral et al. (2025) - CCC FA25

Category: Disease-Motivated Research

- MECP2 dysfunction causes Rett syndrome
- Research on genes linked to neurological disorders
- Goal: understanding neurological dysfunction

Camara et al. (2025) - LU SP25

Category: Basic Metabolic Biology

- Primary focus on amylase gene expression
- Investigates enzyme function
- Disease mentions are contextual

Note: Could fit Disease-Motivated but primary focus is enzyme func

Dehuelbes et al. (2025) - LU SP5

Category: Basic Metabolic Biology

- Focus on energy homeostasis
- Discusses acetyl-CoA and fatty acid metabolism
- Primary goal: metabolic gene coordination

Diaz et al. (2025) - CCC SP25

Category: Basic Cellular Biology

- Focus on immune defense mechanisms
- Investigates V-ATPase genes
- Goal: understanding cellular defense

Ford et al. (2023) - CCC SP23

Category: Basic Cellular Biology

- Focus on insecticide resistance
- Environmental adaptation focus
- Not disease-focused

Gill & Alcazar (2025) - CCC FA25

Category: Disease-Motivated Research

- Discusses mood, anxiety, mental health disorders
- Research linked to mental health
- Goal: understanding psychiatric conditions

Godoy-Pena et al. (2025) - CCC FA25

Category: Basic Cellular Biology

- Focus on sleep regulation genes
- No disease driving research
- Goal: understanding sleep mechanism

Haubelt & Alcazar et al. (2025) - CCC FA25

Category: Disease-Motivated Research

- Title mentions PD Risk (Parkinson's)
- Focus on dopaminergic synapse
- Goal: understanding neuropsychiatric disorders

Henriquez et al. (2025) - COD WI24

Category: Basic Cellular Biology

- Focus on immune pathway expression
- Disease mentions are contextual
- Goal: understanding immune defense

Holmes (2025) - CCC SP25

Category: Basic Cellular Biology

- Focus on melanogenesis and neuromelanin
- Disease mention is for relevance
- Goal: understanding pigment biology

Lemus et al. (2025) - FA25 CCC

Category: Basic Cellular Biology

- Focus on amylase gene expression
- Disease mention is brief
- Goal: understanding digestive enzymes

Logan et al. (2025) - CCC FA25

Category: Disease-Motivated Research

- Introduction on cardiovascular disorders
- Research on blood pressure regulation
- Explicit disease-driven research

Luera et al. (2025) - CCC FA25

Category: Basic Cellular Biology

- Focus on feeding behavior signaling
- No disease driving research
- Goal: understanding neuroendocrine signaling

Meraz et al. (2023) - CCC SP23

Category: Disease-Motivated Research

- Focus on tumor suppressor genes
- Research on cancer-related genes
- Goal: understanding cancer gene expression

Nii et al. (2025) - CCC FA25

Category: Basic Metabolic Biology

- Focus on ACC expression and lipid synthesis
- Disease mentions are applications
- Main goal: iron-dependent metabolic regulation

Otala et al. (2025) - LU SP25

Category: Basic Metabolic Biology

- Focus on fatty acid metabolism genes
- Primary goal: validating fly as model
- Disease mentions are context

Paderna et al. (2025) - CCC SP25

Category: Disease-Motivated Research

- Title mentions Albinism
- Research on OCA4 gene mutations
- Goal: understanding albinism genes

Paramo-Ojeda et al. (2025) - CCC SP25

Category: Disease-Motivated Research

- Title mentions KRAS-Driven Colon Cancer
- Focus on oncogenes and tumor suppressors
- Goal: modeling cancer pathways

Pedireddi et al. (2025) - CCC SP25

Category: Basic Cellular Biology

- Focus on polysaccharide digestion
- No disease driving research
- Goal: understanding digestive enzyme distribution

Rodriguez (2025) - CCC SP25

Category: Disease-Motivated Research

- Introduction on galactosemia
- Focus on GALM gene mutations
- Research motivated by disease

Sakana et al. (2025) - CCC SP25

Category: Other

- Focus on developmental patterning
- No disease or specific biological process focus
- Goal: understanding gene regulation in development

Alternative: Developmental Biology

Trevino et al. (2022) - CCC SP22

Category: Disease-Motivated Research

- Title mentions Zellweger Spectrum Disorder
- Research on peroxisomal disorder genes
- Goal: validating Drosophila as disease model

Tuttle et al. (2025) - CCC SP25

Category: Basic Cellular Biology

- Focus on tryptophan metabolism and aging
- No disease driving research
- Goal: understanding metabolism-aging connection

Zlaket et al. (2023) - CCC SP23

Category: Disease-Motivated Research

- Introduction on Zellweger's Syndrome
- Research on SCP2 gene function
- Focus on disease-related genes